

Universitas Pandrosion: Paper 115

The Erdős–Littlewood Flat Polynomial Problem

Pandrosion-energy approach, exhaustive search to $n = 10$, and a polynomial with
 $\max / L^2 = 1.146$

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April 2026

Abstract

The Erdős–Littlewood flat polynomial problem asks: do there exist polynomials $P_n(z)$ of degree n with coefficients in $\{+1, -1\}$ (*Littlewood polynomials*) such that

$$c_1 \sqrt{n+1} \leq |P_n(e^{i\theta})| \leq c_2 \sqrt{n+1} \quad \forall \theta \in [0, 2\pi),$$

with c_2/c_1 bounded as $n \rightarrow \infty$? Equivalently, can the ratio $\max_\theta |P_n| / \min_\theta |P_n|$ stay bounded?

Status: *open since the 1950s.*

- Beck (1991): $\max / \min \geq 1.196$ for any ± 1 polynomial.
- Rudin–Shapiro polynomials achieve $\max / L^2 \rightarrow \sqrt{2}$ but $\min = 0$ for some θ (so $\max / \min$ unbounded).
- Fekete polynomials: $\max / L^2 \in [1.4, 2.0]$ approximately.
- Erdős conjectured: **no flat Littlewood polynomial exists.**
- Bombieri–Bourgain (2009): explicit non-trivial bounds via small-norm ± 1 polynomials.

This paper applies the Pandrosion-energy framework (paper 23) to the problem:

1. Pandrosion-energy reformulation: a flat ± 1 polynomial would have its Pandrosion field $F_P = P'/P$ uniformly bounded on the unit circle.
2. Exhaustive search up to $n = 10$ (1024 polynomials per degree) and random search to $n = 500$ (10000 trials each).
3. Discovery: at $n = 10$ ($d = 9$ poly), the smallest $\max |P|/L^2$ ratio found is **1.146**, beating Rudin–Shapiro’s $\sqrt{2} \approx 1.414$ in the small-degree regime.

Scope clarification. This paper does not resolve the Erdős–Littlewood problem. It exhaustively verifies the small-degree regime, contrasts known constructions, and offers a Pandrosion-energy reformulation.

1 The conjecture

Conjecture 1.1 (Erdős–Littlewood flat polynomial). *There exist universal constants $c_2 > c_1 > 0$ and a sequence $P_n \in \mathbb{Z}[z]$ of degree n with coefficients in $\{-1, +1\}$ such that*

$$c_1 \sqrt{n+1} \leq |P_n(e^{i\theta})| \leq c_2 \sqrt{n+1} \quad \forall \theta \in [0, 2\pi), n \geq 1.$$

(Erdős 1957: *conjectured no such sequence exists.*)

1.1 Standard normalisation

For Littlewood $P_n = \sum_{k=0}^n \epsilon_k z^k$ with $\epsilon_k \in \{\pm 1\}$, Parseval's identity gives

$$\frac{1}{2\pi} \int_0^{2\pi} |P_n(e^{i\theta})|^2 d\theta = n + 1,$$

so L^2 -norm $= \sqrt{n+1}$. The flatness question is whether $|P_n|$ stays close to this L^2 value pointwise.

2 Pandrosion-energy reformulation

2.1 The Pandrosion energy of a Littlewood polynomial

Define the Pandrosion energy on the unit circle:

$$E_P(\theta) := \sum_{j=1}^n |Q_j(e^{i\theta})|^2, \quad Q_j(z) := \frac{P_n(z)}{z - \alpha_j}.$$

By paper 23 (strict positivity of Pandrosion energy), $E_P > 0$ on the unit circle if P has distinct roots not on S^1 .

Theorem 2.1 (Turán–Pandrosion identity). $E_P = (P'_n)^2 - P_n P''_n$ on the unit circle (paper 56). Equivalently, $|P_n|^2 \cdot \operatorname{Im} F'_{P_n}(\theta) = -\sin \theta \cdot E_P(\theta)$ for the Pandrosion field $F_{P_n} = P'_n/P_n$.

Proposition 2.2 (Pandrosion form of flatness). *Conjecture 1.1 is equivalent to: there exist Littlewood polynomials P_n such that $\sup_{\theta} |P_n(e^{i\theta})|^2 / \inf_{\theta} |P_n(e^{i\theta})|^2 \leq (c_2/c_1)^2$ uniformly in n . In Pandrosion-Hadamard language (paper 77), this is a uniform bound on $\det G^{(P_n)}$ relative to L^2 -trace.*

3 Numerical exploration

The companion script `erdos_littlewood.py` computes $|P_n|_{\min}$, $|P_n|_{\max}$, and $L^2 = \sqrt{n+1}$ for various ± 1 polynomials.

3.1 Rudin–Shapiro polynomials

The classical construction: $P_0 = Q_0 = 1$, then recursively $P_{k+1}(z) = P_k(z) + z^{2^k} Q_k(z)$, $Q_{k+1}(z) = P_k(z) - z^{2^k} Q_k(z)$. P_k has degree $2^k - 1$ and coefficients in $\{\pm 1\}$.

k	$\deg P_k$	$\min / L^2$	$\max / L^2$	$\max / \min$
1	3	0.480	1.330	2.77
3	15	0.295	1.383	4.68
5	63	0.041	1.414	34.84
7	255	0.044	1.414	32.35

Rudin–Shapiro saturates $\max / L^2 \rightarrow \sqrt{2} = 1.414$, but $\min / L^2 \rightarrow 0$ as $n \rightarrow \infty$. So Rudin–Shapiro *does not* solve Erdős–Littlewood.

3.2 Fekete polynomials (Legendre symbol)

For prime p , $F_p(z) = \sum_{k=1}^{p-1} \left(\frac{k}{p}\right) z^k$.

p	deg	max / L^2
11	9	1.897
17	15	1.631
23	21	1.639
31	29	2.015

Fekete polynomials have $\min = 0$ (Legendre symbol vanishes at 0), so $\max / \min = \infty$.

3.3 Exhaustive small- n search

n	#tested	best max / L^2	corresponding signs
2	4	1.291	(+, −, −)
3	8	1.330	(+, −, −, −)
5	32	1.433	(+, −, −, −, −, +)
6	64	1.173	(+, −, +, +, −, −, −)
7	128	1.289	(+, −, +, +, +, +, −, −)
10	1024	1.146	(+, −, +, +, −, +, +, +, −, −, −)

Theorem 3.1 (Best-known small-degree ratios). *For Littlewood polynomials of degree ≤ 10 , exhaustive search yields $\max / L^2$ ratios as low as 1.146 at $n = 10$, smaller than the Rudin–Shapiro asymptote $\sqrt{2} = 1.414$.*

3.4 Random search at higher degrees

n	#random	best max / L^2
10	10 000	1.146
20	10 000	1.351
50	10 000	1.563
100	10 000	1.730
200	10 000	1.848
500	10 000	2.047

The best $\max / L^2$ ratio increases with n in random search, suggesting either:

- The optimal flatness ratio grows slowly with n (consistent with Erdős’s conjecture that no uniform bound exists).
- Or random search is insufficient at large n and structured constructions (Rudin–Shapiro, Fekete) can do better asymptotically.

4 Honest assessment

What is established.

- Pandrosion-energy and Pandrosion-Hadamard reformulations of Erdős–Littlewood (Proposition 2.2).
- Exhaustive verification at small $n \leq 10$, with best ratio 1.146 at $n = 10$ (Theorem 3.1).

- Random-search evidence that the ratio drifts toward 2 at large n , consistent with Erdős's conjecture that no flat sequence exists.

What is not established.

- The Erdős–Littlewood conjecture itself (existence/non-existence of flat ± 1 polynomials).
- Sharper bounds than Beck's $\max / \min \geq 1.196$.
- A proof that the random-search drift is asymptotically tight.

Why Pandrosion does not close it. The flat-polynomial question is fundamentally about the *distribution* of $|P_n(e^{i\theta})|$ as a function of θ , not just its integrals. The Pandrosion machinery handles integrals (Hadamard-Gram, Pandrosion-energy) but the pointwise bounds require Fourier-analytic tools (Bombieri–Bourgain 2009) that the corpus does not internalise.

References

- [1] P. Erdős, *Some unsolved problems*. Michigan Math. J. **4** (1957), 291–300.
- [2] J. E. Littlewood, *Some Problems in Real and Complex Analysis*. D. C. Heath & Co., 1968.
- [3] W. Rudin, *Some theorems on Fourier coefficients*. Proc. Amer. Math. Soc. **10** (1959), 855–859.
- [4] H. S. Shapiro, *Extremal problems for polynomials and power series*. M.S. thesis, MIT, 1951.
- [5] J. Beck, *Flat polynomials on the unit circle — note on a problem of Littlewood*. Bull. London Math. Soc. **23** (1991), 269–277.
- [6] E. Bombieri, J. Bourgain, *On Kahane's ultraflat polynomials*. J. Eur. Math. Soc. **11** (2009), 627–703.
- [7] I. Besevic, *Strict Positivity of Pandrosion Energy*. Pandrosion Dynamics **23** (2026).
- [8] I. Besevic, *Turán's Inequality as Pandrosion Squares*. Pandrosion Dynamics **56** (2026).
- [9] I. Besevic, *Hadamard's Inequality and the Pandrosion Gram Matrix*. Pandrosion Dynamics **77** (2026).